

Artificial Intelligence/ML/DL

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Quiz – 3

1. A linear regression model is trained with gradient descent. The training loss decreases for a few iterations and then starts oscillating and diverging. The MOST likely cause is:

B. The learning rate is too high

2. You train a decision tree to full depth on a dataset and observe 0% training error but 35% test error. Pruning the tree increases training error to 8% and reduces test error to 15%. This is BEST explained as

B. Reducing variance at the cost of a small increase in bias

3. Which statement about Random Forests is FALSE?

C. Random Forests always have lower bias than a single fully-grown decision tree

4. For an SVM with a soft margin, increasing the regularization parameter C (as defined in scikit-learn) typically

B. Narrows the margin and penalizes misclassifications more heavily

5. What does the following scikit-learn code do?

```
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(X_train, y_train)
predictions = model.predict(X_test)
```

B. Fits a linear regression on training data and predicts target values for test data

6. What is the purpose of the following code?

```
from sklearn.cluster import KMeans
model = KMeans(n_clusters=3, random_state=42)
model.fit(X)
labels = model.labels_
```

B. Performs K-Means clustering with 3 clusters and stores labels

7. In the code below, what does the parameter `n_estimators=100` mean?

```
from sklearn.ensemble import RandomForestClassifier  
model = RandomForestClassifier(n_estimators=100)
```

C. The forest will contain 100 decision trees

8. Which of the following correctly imports and creates a Decision Tree classifier in scikit-learn?

C. from sklearn.tree import DecisionTreeClassifier; model =
DecisionTreeClassifier()

11. You have the following labels at a decision tree node (binary classification):

Node: 6 positives, 4 negatives (10 samples total)

A candidate split divides the node as follows:

Left child : 5 positives, 1 negative (6 samples)

Right child: 1 positive, 3 negatives (4 samples)

- (a) Compute the Gini impurity of the parent node.
- (b) Compute the weighted Gini impurity of the children after the split.
- (c) State the Gini gain of this split and briefly argue whether this is a good split.

(a) $\text{Gini}(\text{parent}) = 1 - (0.6^2 + 0.4^2) = 0.48$

(b) $\text{Gini}(\text{left}) \approx 0.278$, $\text{Gini}(\text{right}) = 0.375$

Weighted Gini = $0.6(0.278) + 0.4(0.375) \approx 0.317$

(c) Gini Gain = $0.48 - 0.317 = 0.163$

This is a good split because impurity decreases and child nodes are more pure

12. You fit logistic regression to detect fraud. Fraud is 1% of the data. The model predicts “not fraud” for every input and attains 99% accuracy.

- (a) Explain in 2–3 sentences why accuracy is misleading here.
- (b) Propose TWO concrete remedies (beyond simply “get more data”) and briefly explain how each helps.

(a) Why accuracy is misleading

Accuracy is misleading because the dataset is highly imbalanced (only 1% fraud). A model that predicts “not fraud” for all cases still achieves 99% accuracy but completely fails to detect any actual fraud cases. Therefore, accuracy does not reflect the model’s real performance on the minority (fraud) class.

(b) Two remedies

1. Use better evaluation metrics (Precision, Recall, F1-score, ROC-AUC)

These metrics focus on the minority class and measure how well the model detects fraud, unlike accuracy which is dominated by the majority class.

2. Apply class balancing techniques (e.g., oversampling fraud or using class weights)

Oversampling increases the representation of fraud cases, while class weights penalize misclassification of fraud more heavily. This helps the model learn patterns of the minority class and improves fraud detection

9. Consider a simple linear regression $y = w \cdot x + b$ fit by ordinary least squares on the following four data points:

x: 1, 2, 3, 4

y: 2, 2, 4, 5

(a) Compute the mean of x and the mean of y.

(b) Using the closed-form formula $w = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$, compute w and then b.

(c) Report the predicted value for $x = 5$ and the residual for the point (3, 4).

10. A binary classifier outputs the following probabilities for 5 test points. The true labels are in brackets.

$P(y=1 | x) = [0.9, 0.4, 0.8, 0.3, 0.6]$

$y_{\text{true}} = [1, 0, 0, 1, 1]$

(a) Using threshold 0.5, compute accuracy, precision, recall, and F1-score for class 1.

(b) Give one concrete scenario where you would prefer a threshold of 0.3 over 0.5, and justify.

Answer : 9

Given:- $x = 1, 2, 3, 4$
 $y = 2, 2, 4, 5$

Mean of x and y :-

$$\bar{x} = \frac{1+2+3+4}{4} = \frac{10}{4} = 2.5$$

$$\bar{y} = \frac{2+2+4+5}{4} = \frac{13}{4} = 3.25$$

Compute w and b :-

Step table:-

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$
1	2	-1.5	-1.25	1.875	2.25
2	2	-0.5	-1.25	0.625	0.25
3	4	-0.5	0.75	0.375	0.25
4	5	-1.5	1.75	2.625	2.25

$$w = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = 5.5$$

$$\sum (x_i - \bar{x})^2 = 5$$

$$w = \frac{5.5}{5} = 1.1$$

Intercept:-

$$b = \bar{y} - w\bar{x}$$

$$b = 3.25 - (1.1 \times 2.5) = 3.25 - 2.75 = 0.5$$

Prediction & Residual:-

Prediction for $x = 5$:

$$y = 1.1(5) + 0.5 = 6$$

Residual for $(3, 4)$:

Predicted:

$$y = 1.1(3) + 0.5 = 3.8$$

Residual:

$$= 4 - 3.8 = 0.2$$

Answer 10:

Given:

Probabilities: $[0.9, 0.4, 0.8, 0.3, 0.6]$

True labels: $[1, 0, 0, 1, 1]$

a) Convert to predicted labels
(threshold = 0.5)

$[1, 0, 0, 1, 1]$

Confusion Matrix:-

TP = 2 (1st, 5th)

FP = 1 (3rd)

TN = 1 (2nd)

FN = 1 (4th)

Metrics:-

Accuracy:-

$$= \frac{TP + TN}{\text{Total}} = \frac{2+1}{5} = 0.6$$

$$\text{Precision:-} = \frac{TP}{TP + FP} = \frac{2}{3} \approx 0.67$$

$$\text{Recall:-} = \frac{TP}{TP + FN} = \frac{2}{3} \approx 0.67$$

$$\text{F1-Score:-} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \approx 0.67$$

b) When use threshold = 0.3?

Example: Fraud detection / medical diagnosis

- Lower threshold $\rightarrow$ more positive detected
- Reduces false negatives (missing real fraud)
- Important when missing a positive case is costly.